DO LARGER CRYPTOCURRENCIES EXPERIENCE LOWER PRICE VOLATILITY?

BY: LAURA MELO
ECO 590 – SPRING 2026

RESEARCH

Research Question: Do cryptocurrencies with larger market capitalizations tend to have lower price volatility?

Motivation: In today's financial system, cryptocurrency markets play a crucial role, and their prices are changing frequently. Market capitalization shows the size of the cryptocurrency, while price volatility shows how much the price has changed. This project will research how these two metrics are related to better understand the differences in risk and stability across cryptocurrencies.

Variables Collected:

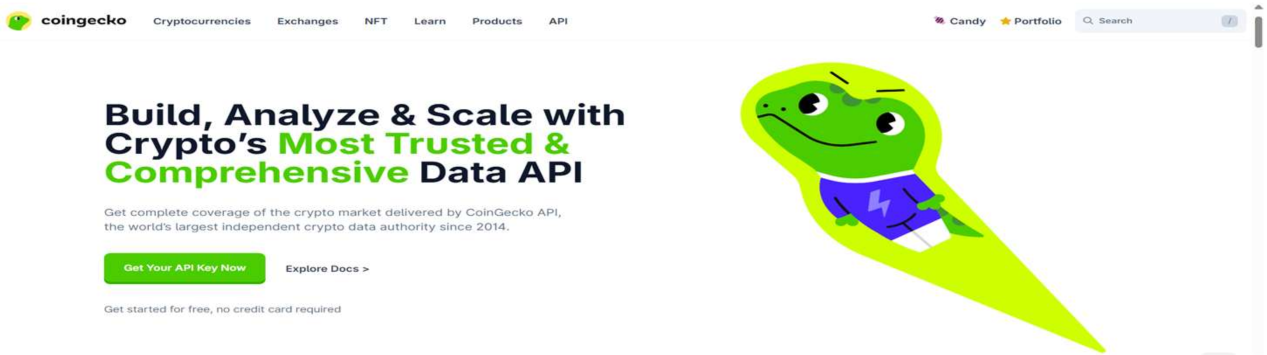
- Cryptocurrency Name
- Current Price
- Market Capitalization
- Trading Volume
- 24-Hour Price Change
- Price Volatility
- Volume to Market Cap Ratio

DATA

Source: CoinGecko API

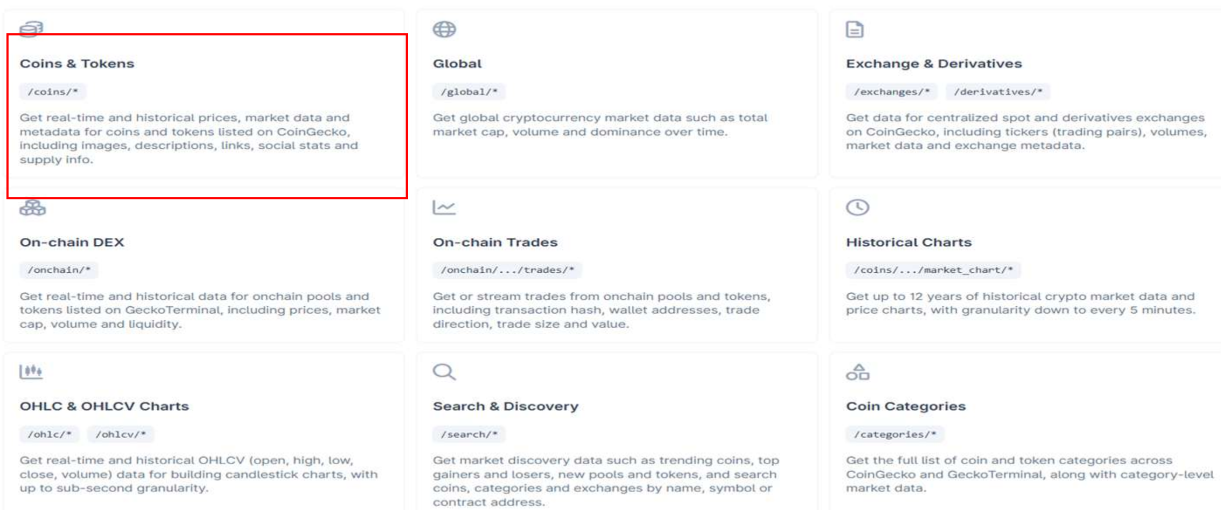
API Endpoint: I used the CoinGecko /coins/markets endpoint, which gave me the current market data for different cryptocurrencies

Difficulty: The hardest part was looking for the correct endpoint because there were various endpoints for different types of data on cryptocurrencies, but once I found the /coins/markets endpoint, the data was well organized



DATA COLLECTION (PART 1)

CoinGecko → Coins & Tokens → /coins/markets



DATA COLLECTION (PART 2)

Looked through the variables & selected the needed ones


```
[
  {
    "id": "bitcoin",
    "symbol": "btc",
    "name": "Bitcoin",
    "image": "https://coin-images.coingecko.com/coins/images/1/large/bitcoin.png?1696501400",
    "current_price": 80208,
    "market_cap": 1605847553832,
    "market_cap_rank": 1,
    "fully_diluted_valuation": 1605847553832,
    "total_volume": 34455920182,
    "high_24h": 82041,
    "low_24h": 79880,
    "price_change_24h": -1345.58070331591,
    "price_change_percentage_24h": -1.64994,
    "market_cap_change_24h": -26406621864.9063,
    "market_cap_change_percentage_24h": -1.6178,
    "circulating_supply": 20028418,
    "total_supply": 20028418,
    "max_supply": 21000000,
    "ath": 126080,
    "ath_change_percentage": -36.38355,
    "ath_date": "2025-10-06T18:57:42.558Z",
    "atl": 67.81,
    "atl_change_percentage": 118184.52513,
    "atl_date": "2013-07-06T00:00:00.000Z",
    "roi": null,
    "last_updated": "2026-05-12T17:16:24.549Z"
  },
```

DATA COLLECTION (PART 3)

Used Python to request data from API

1. Creating the dataset from the API

```
url = "https://api.coingecko.com/api/v3/coins/markets?vs_currency=usd" #link of CoinGecko API (where I get the crypto data)

#settings for the API request
params = {
    "vs_currency": "usd", #get prices in US dollars
    "order": "market_cap_desc", #sort coins by biggest market cap first
    "per_page": 50, #gets the top 50 cryptocurrencies
    "page": 1, #first page of results
    "sparkline": False, #do not need extra graph data
    "price_change_percentage": "24h" #include 24-hour price change %
}
response = requests.get(url, params=params) #sends request to the API and gets the data
data = response.json() #converts the response into JSON format
df = pd.DataFrame(data) #turns the data into a pandas dataframe
```

DATA COLLECTION (PART 4)

Checking dataset was collected correctly

5. Making sure the data set was collected correctly

```
df.info() #checks for column names and data types
df.isnull().sum() #checks if any of the columns have missing values
df.describe() #shows the summary statistics to check if the numbers in the data set make sense
```

```
<class 'pandas.core.frame.DataFrame'>
Index: 49 entries, 0 to 49
Data columns (total 7 columns):
 #   Column                        Non-Null Count  Dtype  
---  ------                        --------------  -----  
 0   name                          49 non-null     object 
 1   current_price                 49 non-null     float64
 2   market_cap                    49 non-null     int64  
 3   total_volume                  49 non-null     float64
 4   price_change_percentage_24h   49 non-null     float64
 5   price_volatility              49 non-null     float64
 6   volume_to_market_cap_ratio    49 non-null     float64
dtypes: float64(5), int64(1), object(1)
memory usage: 3.1+ KB
```

NEW VARIABLES

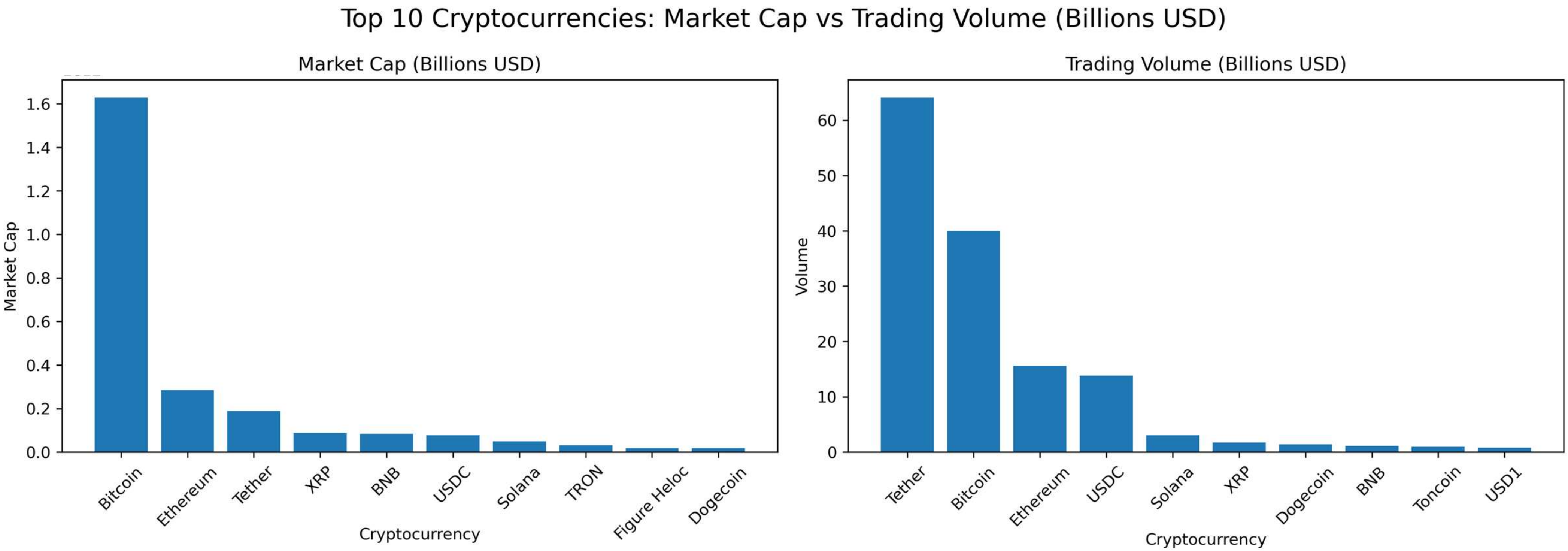
Price Volatility:

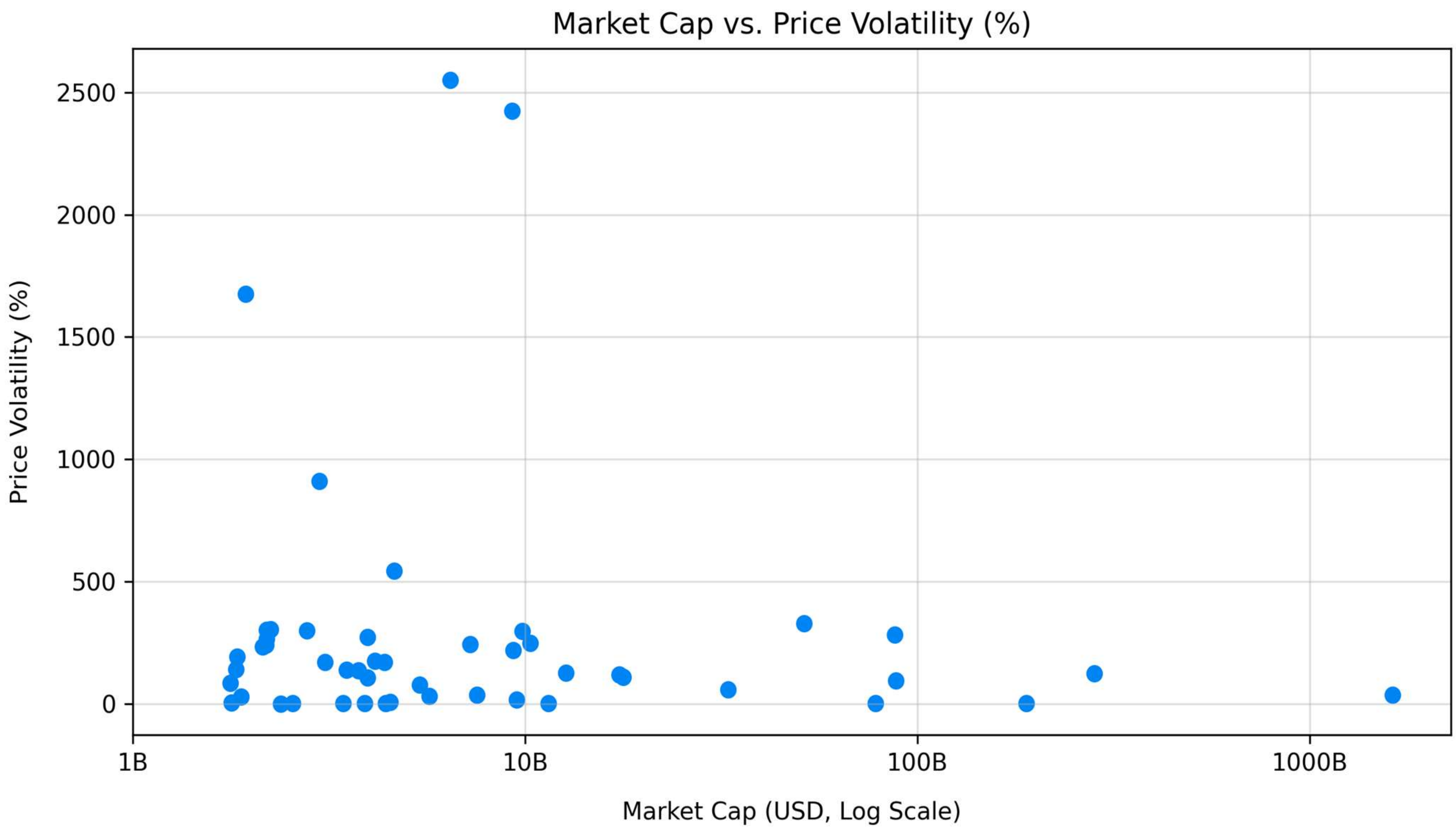
- Measures the absolute value of the 24-hour percent price change
- Helped me focus on how large the price movement was, not whether the price went up or down

Volume to Market Cap Ratio:

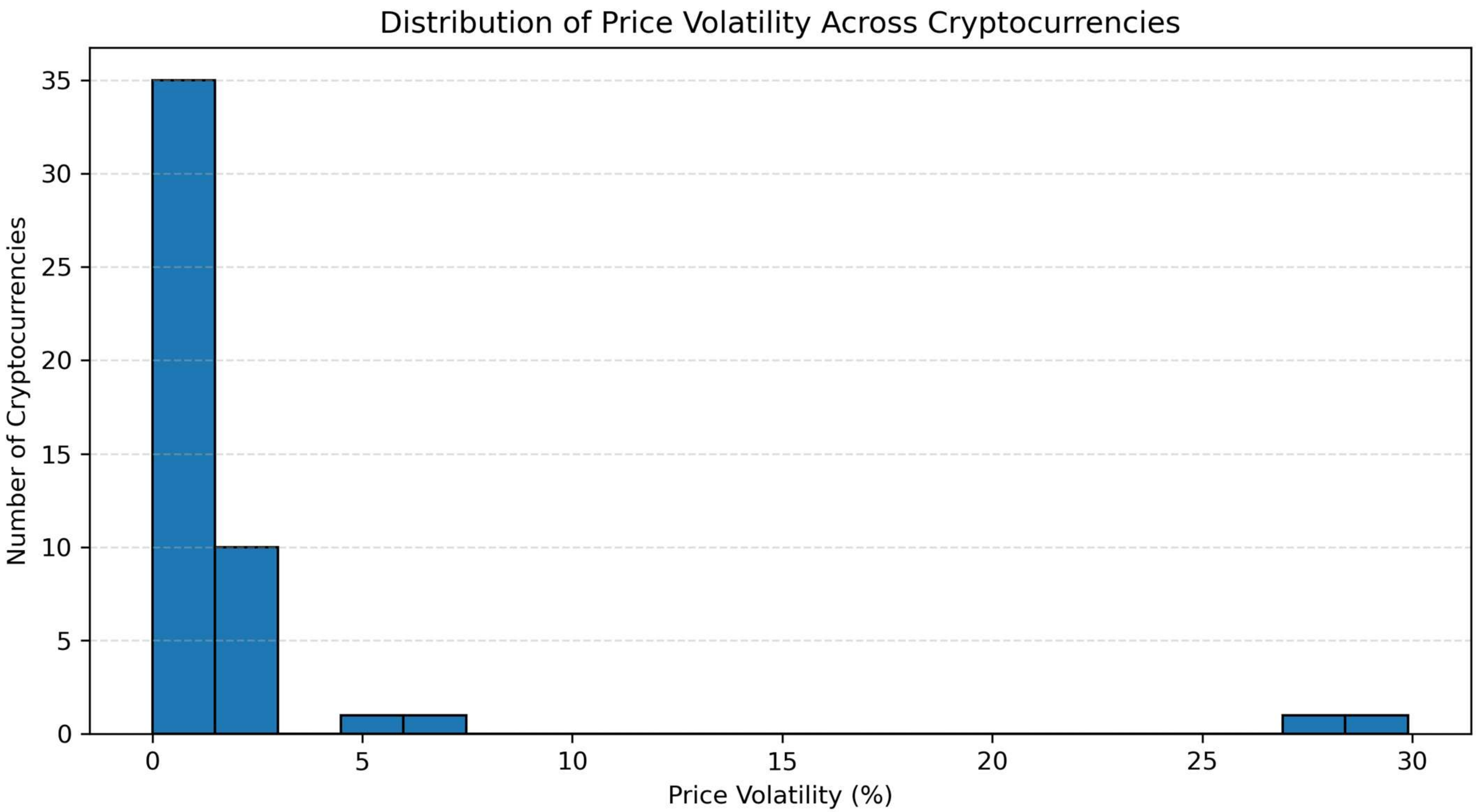
- Divides the trading volume by the market capitalization
- Shows how much trading activity a cryptocurrency has when compared to its overall size
- A higher ratio means that the cryptocurrency is being traded more heavily when compared to its market value

	name	current_price	market_cap	total_volume	price_change_percentage_24h	price_volatility	volume_to_market_cap_ratio
0	Bitcoin	80386.000000	1609870659581	3.459339e+10	-1.46330	1.46330	0.021488
1	Ethereum	2273.850000	274412142391	1.474437e+10	-2.33447	2.33447	0.053731
2	Tether	0.999652	189762732842	5.892913e+10	0.00756	0.00756	0.310541
3	XRP	1.430000	88454428629	2.260380e+09	-2.74507	2.74507	0.025554
4	BNB	654.510000	88221854297	1.129780e+09	-0.78602	0.78602	0.012806
5	USDC	0.999836	77166265384	1.270862e+10	0.01314	0.01314	0.164691
6	Solana	94.400000	54553316828	3.595021e+09	-2.85220	2.85220	0.065899
7	TRON	0.347794	32970233462	4.913829e+08	-0.88628	0.88628	0.014904
8	Figure Heloc	1.032000	18213538966	2.164429e+07	-0.72296	0.72296	0.001188
9	Dogecoin	0.108536	16737659692	1.177740e+09	-2.29127	2.29127	0.070365


Top 10 Cryptocurrencies: Market Cap vs Trading Volume (Billions USD)
Market Cap (Billions USD)
Market Cap
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4
1.6
Bitcoin
Ethereum
Tether
XRP
BNB
USDC
Solana
TRON
Figure Heloc
Dogecoin
Cryptocurrency
Trading Volume (Billions USD)
Volume
0
10
20
30
40
50
60
Tether
Bitcoin
Ethereum
USDC
Solana
XRP
Dogecoin
BNB
Toncoin
USD1
Cryptocurrency



Source: CoinGecko



Source: CoinGecko

MODELS

Model 1 (Research Model):

$\text{Price Volatility}_i=\beta_0+\beta_1\text{Log Trading Volume}_i+\beta_2\text{Log Market Cap}_i+u_i$

Model 2 (Filtered Model):

$\text{Price Volatility}_i=\beta_0+\beta_1\text{Log Trading Volume}_i+\beta_2\text{Log Market Cap}_i+u_i$

Model 3 (Preferred Model):

$\text{Price Volatility}_i=\beta_0+\beta_1\text{Log Market Cap}_i+\beta_2\text{Log Trading Volume}_i+\beta_3\text{Volume-to-Market Cap Ratio}_i+\beta_4\text{Large Crypto}_i+u_i$

Cryptocurrency Regression Results

	Dependent variable:		
	Research Model	Filtered Model	Preferred Model
	(1)	(2)	(3)
Log Trading Volume	1.147**	-0.009	0.407
	(0.428)	(0.225)	(0.609)
Log Market Cap	-1.761**	-0.107	-0.349
	(0.701)	(0.397)	(1.038)
Volume-to-Market-Cap Ratio			19.417
			(12.390)
Large Crypto Dummy			-3.273
			(2.773)
Constant	21.034*	3.859	2.587
	(11.702)	(6.443)	(18.687)
Observations	48	13	48
R2	0.150	0.028	0.224
Adjusted R2	0.112	-0.166	0.152
Residual Std. Error	5.093 (df = 45)	1.149 (df = 10)	4.978 (df = 43)
F Statistic	3.968** (df = 2; 45)	0.144 (df = 2; 10)	3.104** (df = 4; 43)
Note:			*p<0.1; **p<0.05; ***p<0.01

REGRESSION RESULTS

Research Model: $R^2 = 0.150$

- Log Trading Volume = 1.147, significant at 5%
- Log Market Cap = -1.761, significant at 5%

Filtered Model: $R^2 = 0.028$

- Log Trading Volume = -0.009, not significant
- Log Market Cap = -0.107, not significant

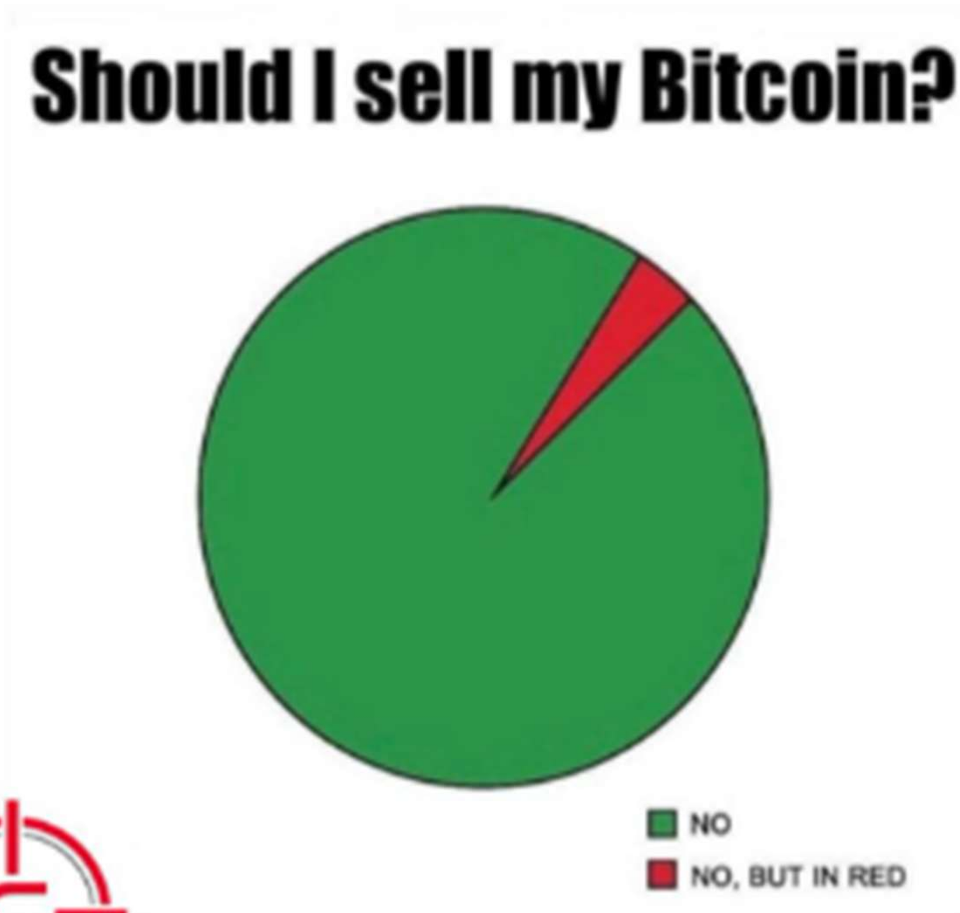
Preferred Model: $R^2 = 0.224$

- Log Trading Volume = 19.417, not significant
- Log Market Cap = 0.407, not significant
- Volume to Market Cap Ratio = -3.273, not significant
- Large Crypto Dummy = -0.349, not significant

CONCLUSION

- The results do not fully support the idea that larger cryptocurrencies always have lower volatility.
- In the preferred regression model, market cap and trading volume were not statistically significant.
- The strongest significant result came from the research model, where higher market cap was linked to lower volatility.
- Overall, cryptocurrency volatility is affected by more than just market size.

THANK YOU


Should I sell my Bitcoin?
NO
NO, BUT IN RED

SOURCES

1. Lecture 4
2. Lecture 6
3. Lecture 8
4. Lecture 9
5. Lecture 10
6. Lecture 11
7. https://www.youtube.com/watch?v=JVQNywo4AbU
8. https://www.youtube.com/watch?v=hpc5jyVpUpw
9. https://stackoverflow.com/questions/61977076/how-to-fetch-data-from-api-using-python
10. https://www.youtube.com/watch?v=NcffDnVBPHA
11. https://www.youtube.com/watch?v=ewXzgCow94g
12. https://htmlcolorcodes.com/
13. https://machinelearningplus.com/plots/python-scatter-plot/
14. https://www.youtube.com/watch?v=xcZib3qAc30
15. https://www.codecademy.com/learn/learn-linear-regression-in-r/modules/linear-regression-in-r/cheatsheet